SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.

2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

*I **G.MOHAMMAD SHAREEF/(192311288)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G.MOHAMMAD SHAREEF

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

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Fundamental concepts in web Development

1. XML (Extensible Markup Language

* XML uses user-defined tags to describe data.
* It separates data from presentation, making the same data usable by different applications.
* It is commonly used for data exchange between web application and systems.

Example:-

```
<student>
    <name>Rahul</name>
    <age>20</age>
</student>
```

2. JSP (Java server page)

JSP is a Java-based server-side technology used to create dynamic web pages.

* A JSP page in proceed on the web server and converted into a servelet.

Example:-

```
<html>
<body>
Welcome, <%=request.getParameter("name")%>
</body>
```

3. MVC Architecture:-

MVC (Model-view-controller) is a software architectural pattern that spearaters an application into three components.

* Model :- Handles data and business logic. such as database operations.

Typical flow:-

user → controller → Model → controller → view → user

4. XSLT :- (Extensible stylesheet Language Transformations)

XSLT is a language used to transform XML documents into formats, such as HTML, XML or text.

For example, an XML document containing student can be transformed into an HTML table display in a browser.

5. XML schema:-

An XML schema (XSD) defines the structure, data types and rules that XML documents must follow.

* which elements are allowed.
* The order of elements.

Role in Web applications:-

XML schema is used to validate XML data, ensuring that data exchanged between applications follow an agreed structure.

6. JSTL (JSP standard Tag Library) :-

JSTL is a collection of standard tags that simplify developmen of psq pages.

* conditional statement
* Loops * Formatting * Accessing data

Example:-

```
<c: for Each var="student" items="${student}">
  ${student.name}
</c:for Each>
```

JSTL makes JSP pages cleaner and easier to maintain by reducing Java code embedded in HTML.

7. PHP (PHP: Hypertext preprocessor)

web application:

* PHP code executes on the server.
* It can process forms and user requests.
* It can connect to databases.
* It can generate HTML dynamically.

Example:

```
<?php
$name = "Rahul";
echo "Welcome, " . $name;
?>
```

PHP provides the server-side logic needed to process requests manage data, interact with databases, and generate dynamic web pages.

(2) XML structure and Validation:

XML (Extensible Markup Language) is used to represent and exchange structured data. It organizes information using - defined tags.

XML:-

```
<student>
    <name>Arun</name>
    <course>MCA</course>
</student>
```

XML must be well-formed, meaning tags are properly nested and closed.

2. JSP and PHP: Server-side processing:

JSP (Java Server Pages) and PHP are technologies for processing requests on the server.

* JSP combines HTML with Java-based server-side logic. It is commonly used with Java components and frameworks.

A typical flow is:-

Browser → Web server → JSP/PHP → Database → JSP/PHP → HTML Respone → Browser

3. MVC architecture: organizing the application:-

MVC (Model - view- controller) separates an application into three parts.

*Model: Handles data cand logic, including database.

*view:- present information to the user, such HTML/JSP pages.

*controller:- Receives user requests, decides what operation is needed, and connects the Model.

For example, when a user requestes information, the controller reciever, the Model.

4. Database connectivity:-

web applications usually need persistent storage for information such a user, products, orders, or student records.

For example:-

The general Process is:-

Application → Database connection → SQL query → Database → Result → Application

The Model layer generally manages the database interactions in a MVC application.

5. Parsers:- Processing structed data:

A parsers reads structured data such as XML and converts into a form that application can process.

*DOM parser:-

Loads the XML document into memory as a tree structure, allowing elements to be accessed modified.

*SAX Parser:- Reads XML sequentially and generates events as encounters elements making it suitable for large documents.

6. How everything is interconnected:-

A complete web application can combine these technologies as follows: